Behavior Poisoning in Cooperative MAPPO

CS 763 Final Project Team

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Key finding: Action-level poisoning can degrade clean MAPPO performance, but damage is seed-sensitive and non-monotonic in poison probability.

1. Motivation and Research Question

In cooperative MARL, agents learn from joint trajectories generated by all teammates. A single compromised teammate can therefore shape the training data seen by the whole team. We study a low-access training-time attack: the adversary changes only one agent's action during rollout collection.

Research question

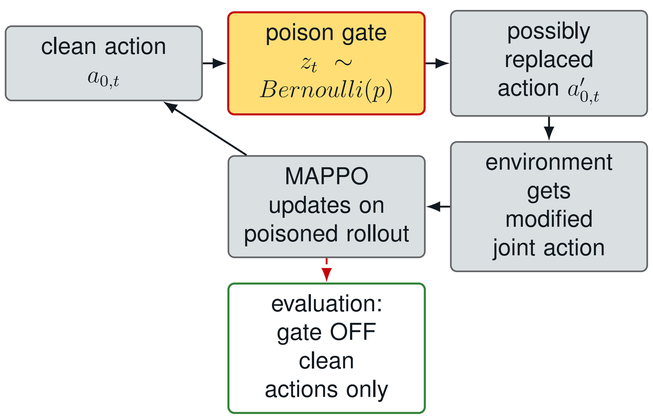
Can action-level behavior poisoning during MAPPO rollout collection degrade final clean-evaluation performance in `simple_spread`?

Task. In `simple_spread`, three agents must cover three landmarks while avoiding unnecessary collisions. Good coordination spreads agents across distinct landmarks; poor coordination causes redundant coverage, missed landmarks, or collisions.

Core idea

training behavior → poisoned rollouts → learned coordination → clean-evaluation drop

2. Threat Model and Attack Variants



$$a'_{0,t} = \begin{cases} \text{poison}(a_{0,t}), & z_t = 1, \\ a_{0,t}, & z_t = 0. \end{cases}$$

Controls	Agent 0 action during training rollout collection only
Cannot change	observations, rewards, parameters, gradients, update rule, or evaluation actions
Goal	maximize $D = R_{clean} - R_{poisoned}$; positive D means poisoning hurt

Attack	Replacement behavior	Intuition
Random	uniform legal action	noisy teammate
Targeted no-op	force action 0	passive teammate
KL-targeted	prefer no-op under KL budget	stealthier intervention

3. Experimental Setup and Metric

- MPE `simple_spread`: 3 agents, 3 landmarks, 25-step episodes.
- Shared-policy recurrent MAPPO with centralized value function.
- 500k training timesteps per condition.
- 32 deterministic clean evaluation episodes per checkpoint.
- Seeds: 7, 13, 21, 31, 37.
- Matched comparison: each poisoned run is compared to the clean run with the same seed.
- $p \in \{0.1, 0.2, 0.3\}$; attack disabled during evaluation.

Reward degradation

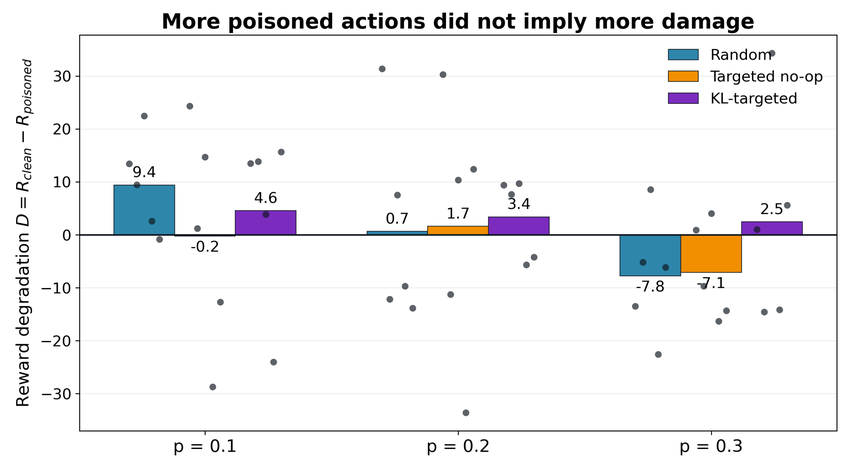
$$D_{seed,attack} = R_{clean,seed} - R_{poisoned,seed}$$

Positive D: poisoning hurt clean evaluation. Negative D: poisoned policy evaluated better.

$$R_t \approx -\sum_{i \in L} \min_i \|x_i - x_\ell\|_2 - \lambda C_t$$

Rewards are usually negative, so values closer to 0 are better. C_t is the collision penalty term.

4. Main Result: More Poisoning Was Not Monotonic



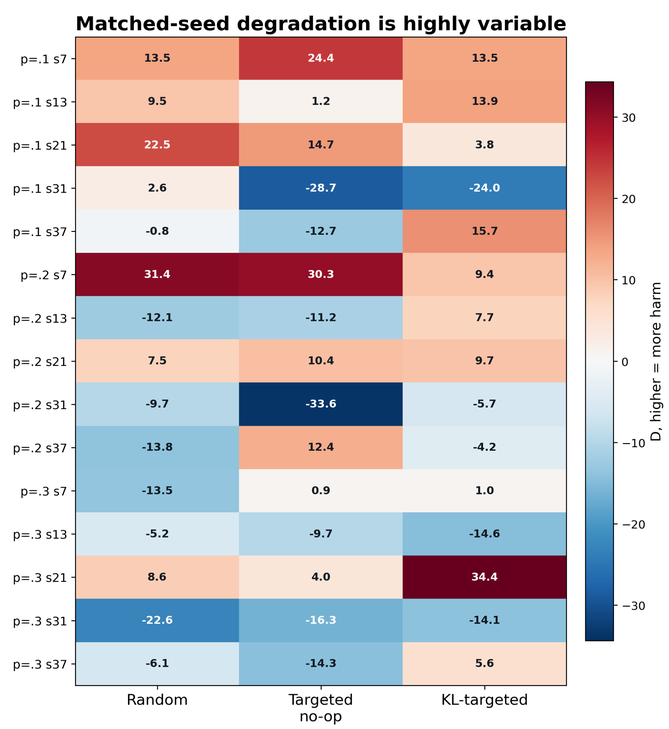
Caption. Positive degradation means poisoning hurt clean evaluation. Increasing p did not consistently increase damage; random and no-op poisoning became beneficial on average at $p = 0.3$.

p	Random	Targeted no-op	KL-targeted	Any hurts	All hurt
0.1	9.43	-0.21	4.58	5/5	3/5
0.2	0.65	1.66	3.39	4/5	2/5
0.3	-7.75	-7.06	2.47	3/5	1/5

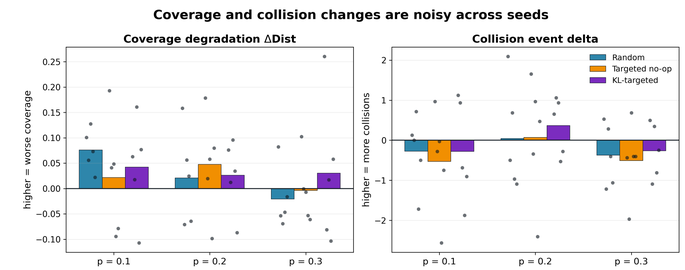
Any hurts = at least one attack has $D > 0$ for that seed. All hurt = all three attacks have $D > 0$ for that seed.

Interpretation. Larger p creates more poisoned actions, but the intervention can also behave like exploration or regularization. KL-targeted poisoning remains harmful on average at all tested p values, but the effect is strongly seed-dependent.

5. Seed Sensitivity



6. Mechanism: Coverage Effects Are Noisy



Caption. Coverage degradation and collision deltas overlap across attacks and p values, helping explain why reward degradation is noisy rather than purely monotonic.

7. Hypothesis Check

Hypothesis	Outcome	Evidence
H1 Clean MAPPO learns coordination	Supported	Clean baseline reaches reasonable coverage/reward
H2 Poisoning hurts clean evaluation	Partially supported	Several attack/p settings have positive D
H3 No-op beats random	Not supported	Targeted no-op is not reliably more damaging
H4 KL-constrained attack can hurt	Partially supported	KL-targeted has positive mean D at all tested p
H5 More poison hurts more	Not supported	$p = 0.3$ improves random/no-op on average

The broad vulnerability claim is supported; monotonic and targeted-dominance claims are not.

8. Takeaways, Limits, and Next Steps

Takeaways

- Behavior poisoning is a real train-time vulnerability: one compromised agent can change the final learned team policy.
- The attack is low-access: no reward, observation, gradient, weight, or evaluation tampering is needed.
- Damage is seed-sensitive and non-monotonic; more poisoned actions do not automatically mean more damage.

Limits: one environment; one compromised agent; $p \in \{0.1, 0.2, 0.3\}$; one no-op target action; five seeds.

Next steps: adaptive target-action search; more seeds and environments; multiple compromised agents; action anomaly detection; robust rollout filtering.

Refs: Yu et al. 2022; Schulman et al. 2017; Lowe et al. 2017; Terry et al. 2021; Mordatch and Abbeel 2018.